借助NetApp文件对象双处理能力和AWS SageMaker实现云数据管理

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借助NetApp文件对象双处理能力和AWS SageMaker实现云数据管理

TR-4967：《使用NetApp文件对象双处理和AWS SageMaker进行云数据管理》

NetApp 公司 Karthikeyan Nagalingam

数据科学家和工程师通常需要访问以NFS格式存储的数据、但在AWS SageMaker中直接从S3协议访问此数据可能会面临挑战、因为AWS仅支持S3存储分段访问。但是、NetApp ONTAP 通过为NFS和S3启用双协议访问来提供解决方案。借助此解决方案 、数据科学家和工程师可以通过NetApp Cloud Volumes ONTAP 的S3存储分段从AWS SageMaker笔记本电脑访问NFS数据。这种方法可以轻松地从NFS和S3访问和共享相同的数据、而无需额外的软件。

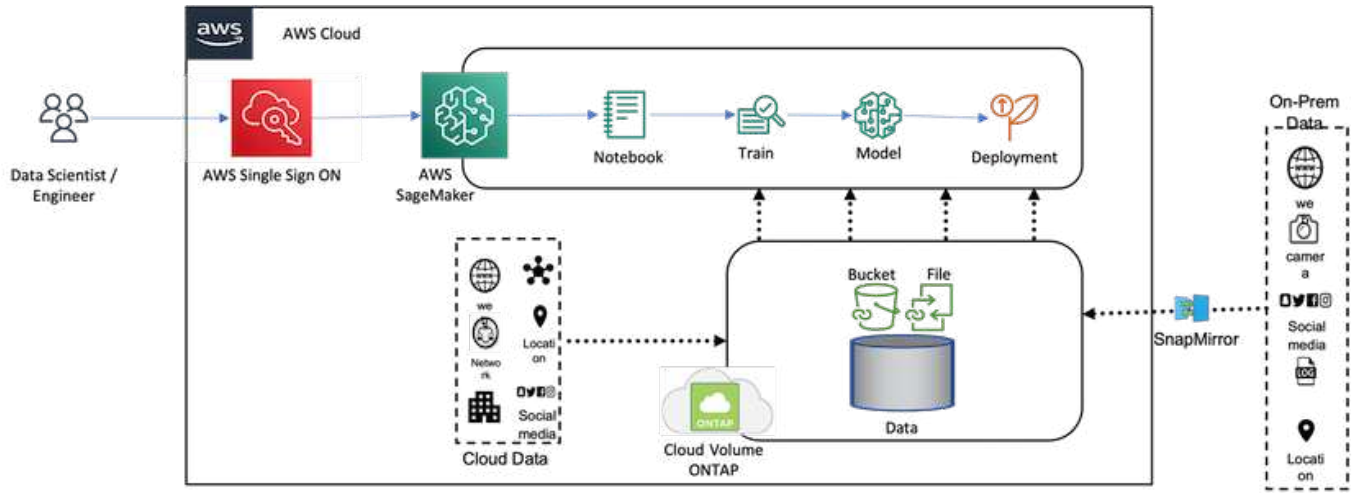
解决方案技术

此解决方案 利用以下技术：

- * AWS SageMaker Notebook.*为开发人员和数据科学家提供机器学习功能、帮助他们高效地创建、训练和部署高质量的ML模型。
- * NetApp BlueXp。*支持发现、部署和操作内部以及AWS、Azure和Google Cloud上的存储。它可以防止数据丢失、网络威胁和计划外中断、并优化数据存储和基础架构。
- * NetApp Cloud Volumes ONTAP。*在AWS、Azure和Google Cloud上提供采用NFS、SMB/CIFS、iSCSI和S3协议的企业级存储卷、使用户可以更灵活地访问和管理云中的数据。

NetApp Cloud Volumes ONTAP 是从BlueXP创建的、用于存储ML数据。

下图显示了解决方案的技术组件。



使用情形摘要

NFS和S3的双协议访问的一个潜在用例是机器学习和数据科学领域。例如、数据科学家团队可能正在使用AWS SageMaker执行机器学习项目、该项目要求访问以NFS格式存储的数据。但是、可能还需要通过S3存储分段访问和共享数据、以便与其他团队成员协作或与使用S3的其他应用程序集成。

通过使用NetApp Cloud Volumes ONTAP 、该团队可以将数据存储在一个位置、并可通过NFS和S3协议访问这些数据。数据科学家可以直接从AWS SageMaker访问NFS格式的数据、而其他团队成员或应用程序则可以通过S3存储分段访问相同的数据。

这种方法可以轻松高效地访问和共享数据、而无需在不同的存储解决方案之间进行额外的软件或数据迁移。此外、它还可以简化工作流并在团队成员之间进行协作、从而更快、更有效地开发机器学习模型。

数据科学家和其他应用程序的数据双重性

数据在NFS中可用、并可从AWS SageMaker的S3访问。

技术要求

您需要使用NetApp BlueXP、NetApp Cloud Volumes ONTAP 和AWS SageMaker笔记本电脑作为数据双用用例。

软件要求

下表列出了实施此用例所需的软件组件。

软件	数量
BlueXP	1.
NetApp Cloud Volumes ONTAP	1.
AWS SageMaker笔记本电脑	1.

部署过程

部署数据双率解决方案 涉及以下任务：

- BlueXP连接器
- NetApp Cloud Volumes ONTAP
- 用于机器学习的数据
- AWS SageMaker
- 通过Jupyter笔记本电脑经验证的机器学习

BlueXP连接器

在此验证中、我们使用了AWS。它也适用于Azure和Google Cloud。要在AWS中创建BlueXP Connector、请完成以下步骤：

1. 我们根据BlueXP中的mcarl-marketplace-subscription使用了这些凭据。

2. 选择适合您环境的区域(例如us-east-1 [N.Virginia)、然后选择身份验证方法(例如、假设角色或AWS密钥)。在此验证中、我们将使用AWS密钥。
3. 提供连接器的名称并创建角色。
4. 根据您是否需要公共IP、提供VPC、子网或密钥对等网络详细信息。
5. 提供安全组的详细信息、例如从源类型访问HTTP、HTTPS或SSH、例如Anywhere和IP范围信息。
6. 查看并创建BlueXP Connector。
7. 确认在AWS控制台中运行的是BlueXP EC2实例状态、然后从*网络连接*选项卡中检查IP地址。
8. 从BlueXP门户登录到连接器用户界面、或者您也可以使用IP地址从浏览器进行访问。

NetApp Cloud Volumes ONTAP

要在BlueXP中创建Cloud Volumes ONTAP 实例、请完成以下步骤：

1. 创建新的工作环境、选择云提供商、然后选择Cloud Volumes ONTAP 实例的类型(例如、适用于ONTAP 的单CVO、HA或Amazon FSxN)。
2. 提供Cloud Volumes ONTAP 集群名称和凭据等详细信息。在此验证中、我们创建了一个名为的Cloud Volumes ONTAP 实例 `svm_sagemaker_cvo_sn1`。
3. 选择Cloud Volumes ONTAP 所需的服务。在此验证中、我们选择仅监控、因此禁用了*数据感知与合规性*和*备份到云服务*。
4. 在*位置和连接*部分中、选择AWS区域、VPC、子网、安全组、SSH身份验证方法、 以及密码或密钥对。
5. 选择充电方法。我们使用*专业版*进行此验证。
6. 您可以选择预配置的软件包、例如* POC和小型工作负载*、数据库和应用程序数据生产工作负载、经济高效的灾难恢复*或*最高性能生产工作负载。在此验证中、我们选择了* POC和小型工作负载*。
7. 创建具有特定大小、允许的协议和导出选项的卷。在此验证中、我们创建了一个名为的卷 `vol1`。
8. 选择配置文件磁盘类型和分层策略。在此验证中、我们禁用了*存储效率*和*通用SSD–动态性能*。
9. 最后、查看并创建Cloud Volumes ONTAP 实例。然后等待15到20分钟、让BlueXP创建Cloud Volumes ONTAP 工作环境。
10. 配置以下参数以启用双率协议。ONTAP 9支持双度协议(NFS/S3)。12：1及更高版本。
    a. 在此验证中、我们创建了一个名为的SVM `svm_sagemaker_cvo_sn1` 和卷 `vol1`。
    b. 验证SVM是否支持NFS和S3协议。如果不支持、请修改SVM以支持它们。

```
sagemaker_cvo_sn1::> vserver show -vserver svm_sagemaker_cvo_sn1
                                    Vserver: svm_sagemaker_cvo_sn1
                               Vserver Type: data
                            Vserver Subtype: default
                               Vserver UUID: 911065dd-a8bc-11ed-bc24-
e1c0f00ad86b
                                Root Volume:
svm_sagemaker_cvo_sn1_root
                                  Aggregate: aggr1
                                 NIS Domain: -
                  Root Volume Security Style: unix
                                LDAP Client: -
               Default Volume Language Code: C.UTF-8
                            Snapshot Policy: default
                              Data Services: data-cifs, data-
flexcache,
                                             data-iscsi, data-nfs,
                                             data-nvme-tcp
                                     Comment:
                               Quota Policy: default
                List of Aggregates Assigned: aggr1
 Limit on Maximum Number of Volumes allowed: unlimited
                        Vserver Admin State: running
                  Vserver Operational State: running
   Vserver Operational State Stopped Reason: -
                          Allowed Protocols: nfs, cifs, fcp, iscsi,
ndmp, s3
                       Disallowed Protocols: nvme
            Is Vserver with Infinite Volume: false
                           QoS Policy Group: -
                        Caching Policy Name: -
                                Config Lock: false
                               IPspace Name: Default
                         Foreground Process: -
                    Logical Space Reporting: true
                  Logical Space Enforcement: false
Default Anti_ransomware State of the Vserver's Volumes: disabled
            Enable Analytics on New Volumes: false
    Enable Activity Tracking on New Volumes: false

sagemaker_cvo_sn1::>
```

11. 根据需要创建并安装CA证书。
12. 创建服务数据策略。

```
sagemaker_cvo_sn1::*> network interface service-policy create -vserver
svm_sagemaker_cvo_sn1 -policy sagemaker_s3_nfs_policy -services data-
core,data-s3-server,data-nfs,data-flexcache
sagemaker_cvo_sn1::*> network interface create -vserver
svm_sagemaker_cvo_sn1 -lif svm_sagemaker_cvo_sn1_s3_lif -service-policy
sagemaker_s3_nfs_policy -home-node sagemaker_cvo_sn1-01 -address
172.30.10.41 -netmask 255.255.255.192

Warning: The configured failover-group has no valid failover targets for
the LIF's failover-policy. To view the failover targets for a LIF, use
         the "network interface show -failover" command.

sagemaker_cvo_sn1::*>
sagemaker_cvo_sn1::*> network interface show
Logical    Status     Network            Current       Current Is
Vserver     Interface  Admin/Oper Address/Mask       Node          Port
Home
----------- ---------- ---------- ------------------ -------------
------- ----
sagemaker_cvo_sn1
            cluster-mgmt up/up    172.30.10.40/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            intercluster up/up    172.30.10.48/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            sagemaker_cvo_sn1-01_mgmt1
                         up/up    172.30.10.58/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
svm_sagemaker_cvo_sn1
            svm_sagemaker_cvo_sn1_data_lif
                         up/up    172.30.10.23/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            svm_sagemaker_cvo_sn1_mgmt_lif
                         up/up    172.30.10.32/26    sagemaker_cvo_sn1-
01
                                                                    e0a
true
            svm_sagemaker_cvo_sn1_s3_lif
                         up/up    172.30.10.41/26    sagemaker_cvo_sn1-
```

```
01
                                                                    e0a
true
6 entries were displayed.

sagemaker_cvo_sn1::*>
sagemaker_cvo_sn1::*> vserver object-store-server create -vserver
svm_sagemaker_cvo_sn1  -is-http-enabled true -object-store-server
svm_sagemaker_cvo_s3_sn1 -is-https-enabled false
sagemaker_cvo_sn1::*> vserver object-store-server show

Vserver: svm_sagemaker_cvo_sn1

           Object Store Server Name: svm_sagemaker_cvo_s3_sn1
               Administrative State: up
                       HTTP Enabled: true
             Listener Port For HTTP: 80
                      HTTPS Enabled: false
     Secure Listener Port For HTTPS: 443
  Certificate for HTTPS Connections: -
                  Default UNIX User: pcuser
               Default Windows User: -
                            Comment:

sagemaker_cvo_sn1::*>
```

13. 检查聚合详细信息。

```
sagemaker_cvo_sn1::*> aggr show


Aggregate     Size Available Used% State   #Vols  Nodes            RAID
Status
--------- -------- --------- ----- ------- ------ ----------------
------------
aggr0_sagemaker_cvo_sn1_01
           124.0GB   50.88GB   59% online       1 sagemaker_cvo_
raid0,
                                                   sn1-01
normal
aggr1      907.1GB   904.9GB    0% online       2 sagemaker_cvo_
raid0,
                                                   sn1-01
normal
2 entries were displayed.

sagemaker_cvo_sn1::*>
```

14. 创建用户和组。

```
sagemaker_cvo_sn1::*> vserver object-store-server user create -vserver
svm_sagemaker_cvo_sn1 -user s3user

sagemaker_cvo_sn1::*> vserver object-store-server user show
Vserver     User            ID        Access Key          Secret Key
----------- --------------- --------- -------------------
-------------------
svm_sagemaker_cvo_sn1
            root            0         -                   -
   Comment: Root User
svm_sagemaker_cvo_sn1
            s3user          1         0ZNAX21JW5Q8AP80CQ2E

PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr
2 entries were displayed.

sagemaker_cvo_sn1::*>


sagemaker_cvo_sn1::*> vserver object-store-server group create -name
s3group -users s3user -comment ""

sagemaker_cvo_sn1::*>
sagemaker_cvo_sn1::*> vserver object-store-server group delete -gid 1
-vserver svm_sagemaker_cvo_sn1

sagemaker_cvo_sn1::*> vserver object-store-server group create -name
s3group -users s3user -comment "" -policies FullAccess

sagemaker_cvo_sn1::*>
```

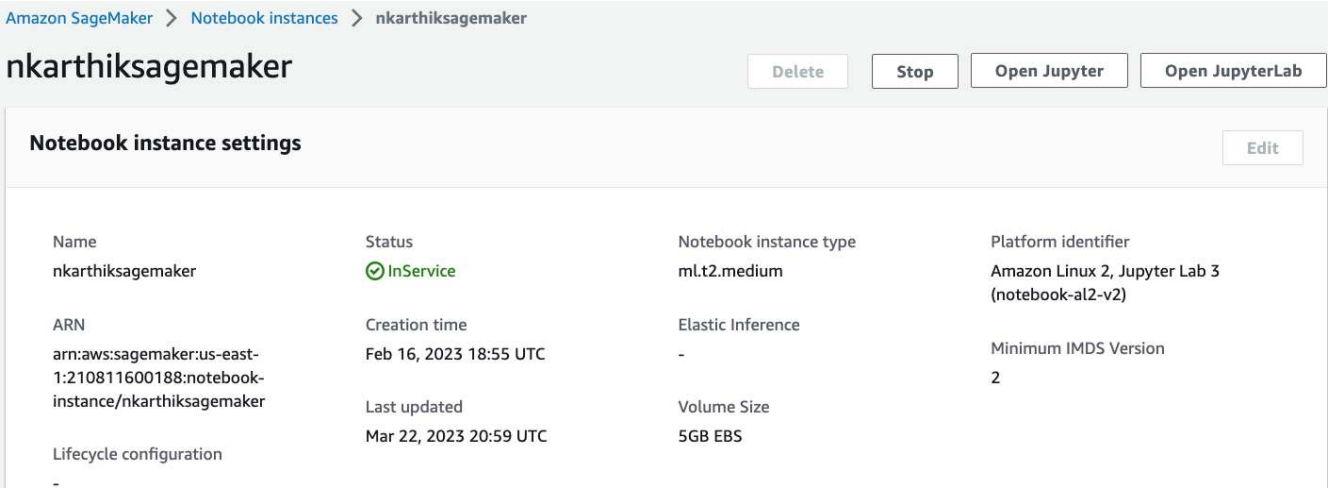
15. 在NFS卷上创建存储分段。

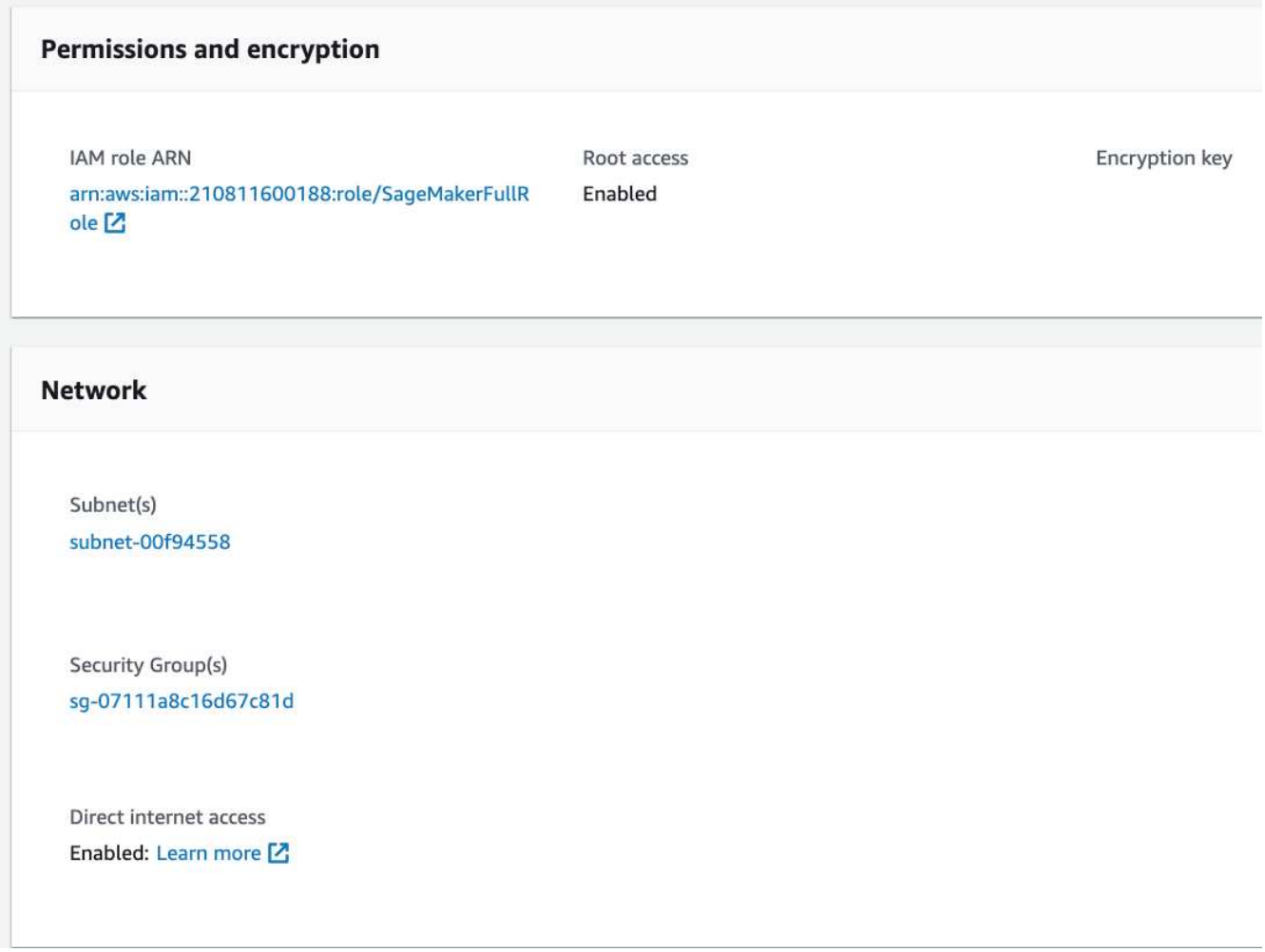
```
sagemaker_cvo_sn1::*> vserver object-store-server bucket create -bucket
ontapbucket1 -type nas -comment "" -vserver svm_sagemaker_cvo_sn1 -nas
-path /vol1
sagemaker_cvo_sn1::*> vserver object-store-server bucket show
Vserver     Bucket          Type     Volume            Size
Encryption Role       NAS Path
----------- --------------- -------- ----------------- ----------
---------- ---------- ----------
svm_sagemaker_cvo_sn1
            ontapbucket1    nas      vol1              -          false
-          /vol1
sagemaker_cvo_sn1::*>
```

AWS SageMaker

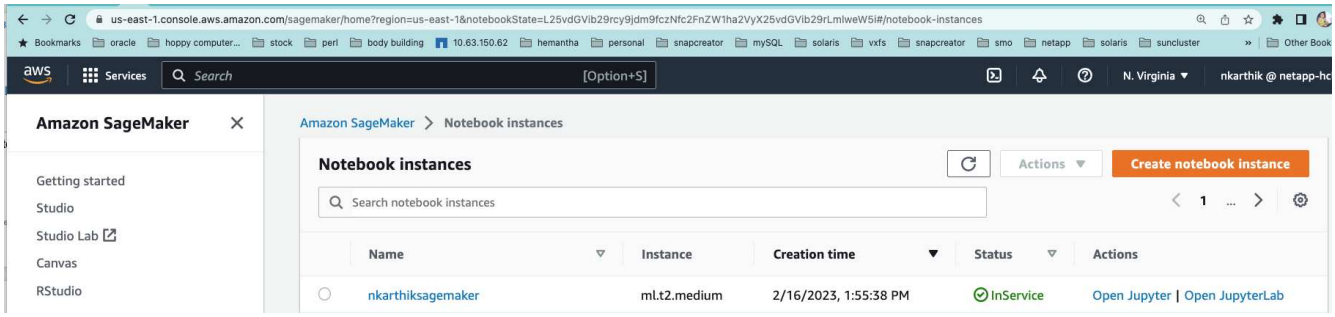
要从AWS SageMaker创建AWS笔记本电脑、请完成以下步骤：

1. 确保正在创建笔记本实例的用户具有AmazonSageMakerFullAccess IAM策略或属于具有AmazonSageMakerFullAccess权限的现有组。在此验证中、用户属于现有组。
2. 请提供以下信息：
   - 笔记本实例名称。
   - 实例类型。
   - 平台标识符。
   - 选择具有AmazonSageMakerFullAccess权限的IAM角色。
   - root访问—启用。
   - 加密密钥-不选择自定义加密。
   - 保留其余默认选项。
3. 在此验证中、SageMaker实例详细信息如下所示：

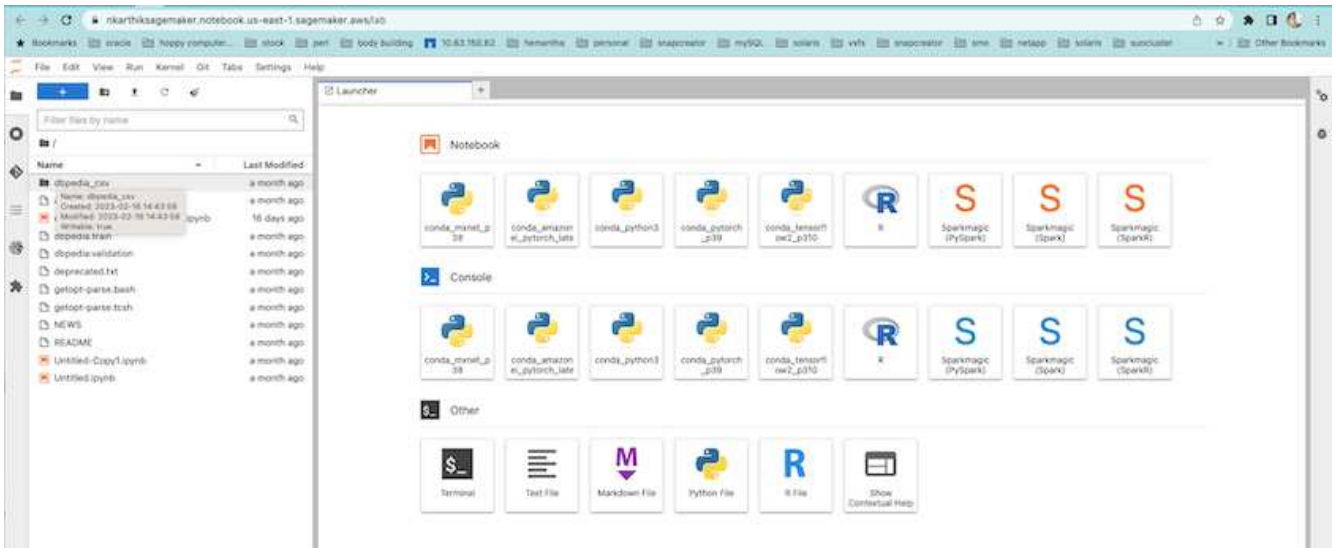




4. 启动AWS笔记本电脑。



5. 打开Jupyter实验室。



6. 登录到终端并挂载Cloud Volumes ONTAP 卷。

```
sh-4.2$ sudo mkdir /vol1; sudo mount -t nfs 172.30.10.41:/vol1 /vol1
sh-4.2$ df -h
Filesystem          Size  Used Avail Use% Mounted on
devtmpfs            2.0G     0  2.0G   0% /dev
tmpfs               2.0G     0  2.0G   0% /dev/shm
tmpfs               2.0G  624K  2.0G   1% /run
tmpfs               2.0G     0  2.0G   0% /sys/fs/cgroup
/dev/xvda1          140G  114G   27G  82% /
/dev/xvdf           4.8G   72K  4.6G   1% /home/ec2-user/SageMaker
tmpfs               393M     0  393M   0% /run/user/1001
tmpfs               393M     0  393M   0% /run/user/1002
tmpfs               393M     0  393M   0% /run/user/1000
172.30.10.41:/vol1  973M  189M  785M  20% /vol1
sh-4.2$
```

7. 使用AWS命令行界面命令检查在Cloud Volumes ONTAP 卷上创建的存储分段。

```
sh-4.2$ aws configure --profile netapp
AWS Access Key ID [None]: 0ZNAX21JW5Q8AP80CQ2E
AWS Secret Access Key [None]: PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr
Default region name [None]: us-east-1
Default output format [None]:
sh-4.2$

sh-4.2$ aws s3 ls --profile netapp --endpoint-url
2023-02-10 17:59:48 ontapbucket1

sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/


2023-02-10 18:46:44       4747 1
2023-02-10 18:48:32         96 setup.cfg

sh-4.2$
```

用于机器学习的数据

在此次验证中、我们使用了一个来自大众社区的DBipedia的数据集、从各种Wikimedia项目中创建的信息中提取结构化内容。

1. 从DBipedia GitHub位置下载数据并提取数据。请使用上一节中使用的相同终端。

```
sh-4.2$ wget
--2023-02-14 23:12:11--
Resolving github.com (github.com)... 140.82.113.3
Connecting to github.com (github.com)|140.82.113.3|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location:  [following]
--2023-02-14 23:12:11--
Resolving raw.githubusercontent.com (raw.githubusercontent.com)...
185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com
(raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 68431223 (65M) [application/octet-stream]
Saving to: ‘dbpedia_csv.tar.gz’

100%[===================================================================
===================================================================
===================>] 68,431,223  56.2MB/s   in 1.2s

2023-02-14 23:12:13 (56.2 MB/s) - ‘dbpedia_csv.tar.gz’ saved
[68431223/68431223]

sh-4.2$ tar -zxvf dbpedia_csv.tar.gz
dbpedia_csv/
dbpedia_csv/test.csv
dbpedia_csv/classes.txt
dbpedia_csv/train.csv
dbpedia_csv/readme.txt
sh-4.2$
```

2. 将数据复制到Cloud Volumes ONTAP 位置、并使用AWS命令行界面从S3存储分段中进行检查。

```
sh-4.2$ df -h
Filesystem          Size  Used Avail Use% Mounted on
devtmpfs            2.0G     0  2.0G   0% /dev
tmpfs               2.0G     0  2.0G   0% /dev/shm
tmpfs               2.0G  628K  2.0G   1% /run
tmpfs               2.0G     0  2.0G   0% /sys/fs/cgroup
/dev/xvda1          140G  114G   27G  82% /
/dev/xvdf           4.8G   52K  4.6G   1% /home/ec2-user/SageMaker
tmpfs               393M     0  393M   0% /run/user/1002
tmpfs               393M     0  393M   0% /run/user/1001
tmpfs               393M     0  393M   0% /run/user/1000
172.30.10.41:/vol1  973M  384K  973M   1% /vol1
sh-4.2$ pwd
/home/ec2-user
sh-4.2$ cp -ra dbpedia_csv /vol1
sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/
                           PRE dbpedia_csv/
2023-02-10 18:46:44       4747 1
2023-02-10 18:48:32         96 setup.cfg
sh-4.2$
```

3. 执行基本验证以确保读取/写入功能在S3存储分段上正常工作。

```
sh-4.2$ aws s3 cp  --profile netapp --endpoint-url  /usr/share/doc/util-
linux-2.30.2 s3://ontapbucket1/ --recursive
upload: ../../../usr/share/doc/util-linux-2.30.2/deprecated.txt to
s3://ontapbucket1/deprecated.txt
upload: ../../../usr/share/doc/util-linux-2.30.2/getopt-parse.bash to
s3://ontapbucket1/getopt-parse.bash
upload: ../../../usr/share/doc/util-linux-2.30.2/README to
s3://ontapbucket1/README
upload: ../../../usr/share/doc/util-linux-2.30.2/getopt-parse.tcsh to
s3://ontapbucket1/getopt-parse.tcsh
upload: ../../../usr/share/doc/util-linux-2.30.2/AUTHORS to
s3://ontapbucket1/AUTHORS
upload: ../../../usr/share/doc/util-linux-2.30.2/NEWS to
s3://ontapbucket1/NEWS
sh-4.2$ aws s3 ls --profile netapp --endpoint-url
s3://ontapbucket1/s3://ontapbucket1/

An error occurred (InternalError) when calling the ListObjectsV2
operation: We encountered an internal error. Please try again.
sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/
                           PRE dbpedia_csv/
```

```
2023-02-16 19:19:27      26774 AUTHORS
2023-02-16 19:19:27      72727 NEWS
2023-02-16 19:19:27       4493 README
2023-02-16 19:19:27       2825 deprecated.txt
2023-02-16 19:19:27       1590 getopt-parse.bash
2023-02-16 19:19:27       2245 getopt-parse.tcsh
sh-4.2$ ls -ltr /vol1
total 132
drwxrwxr-x 2 ec2-user ec2-user  4096 Mar 29  2015 dbpedia_csv
-rw-r--r-- 1 nobody   nobody    2245 Apr 10 17:37 getopt-parse.tcsh
-rw-r--r-- 1 nobody   nobody    2825 Apr 10 17:37 deprecated.txt
-rw-r--r-- 1 nobody   nobody    4493 Apr 10 17:37 README
-rw-r--r-- 1 nobody   nobody    1590 Apr 10 17:37 getopt-parse.bash
-rw-r--r-- 1 nobody   nobody   26774 Apr 10 17:37 AUTHORS
-rw-r--r-- 1 nobody   nobody   72727 Apr 10 17:37 NEWS
sh-4.2$ ls -ltr /vol1/dbpedia_csv/
total 192104
-rw------- 1 ec2-user ec2-user 174148970 Mar 28  2015 train.csv
-rw------- 1 ec2-user ec2-user  21775285 Mar 28  2015 test.csv
-rw------- 1 ec2-user ec2-user       146 Mar 28  2015 classes.txt
-rw-rw-r-- 1 ec2-user ec2-user      1758 Mar 29  2015 readme.txt
sh-4.2$ chmod -R 777 /vol1/dbpedia_csv
sh-4.2$ ls -ltr /vol1/dbpedia_csv/
total 192104
-rwxrwxrwx 1 ec2-user ec2-user 174148970 Mar 28  2015 train.csv
-rwxrwxrwx 1 ec2-user ec2-user  21775285 Mar 28  2015 test.csv
-rwxrwxrwx 1 ec2-user ec2-user       146 Mar 28  2015 classes.txt
-rwxrwxrwx 1 ec2-user ec2-user      1758 Mar 29  2015 readme.txt
sh-4.2$ aws s3 cp --profile netapp --endpoint-url http://172.30.2.248/
s3://ontapbucket1/ /tmp --recursive
download: s3://ontapbucket1/AUTHORS to ../../tmp/AUTHORS
download: s3://ontapbucket1/README to ../../tmp/README
download: s3://ontapbucket1/NEWS to ../../tmp/NEWS
download: s3://ontapbucket1/dbpedia_csv/classes.txt to
../../tmp/dbpedia_csv/classes.txt
download: s3://ontapbucket1/dbpedia_csv/readme.txt to
../../tmp/dbpedia_csv/readme.txt
download: s3://ontapbucket1/deprecated.txt to ../../tmp/deprecated.txt
download: s3://ontapbucket1/getopt-parse.bash to ../../tmp/getopt-
parse.bash
download: s3://ontapbucket1/getopt-parse.tcsh to ../../tmp/getopt-
parse.tcsh
download: s3://ontapbucket1/dbpedia_csv/test.csv to
../../tmp/dbpedia_csv/test.csv
download: s3://ontapbucket1/dbpedia_csv/train.csv to
../../tmp/dbpedia_csv/train.csv
```

```
sh-4.2$
sh-4.2$ aws s3 ls --profile netapp --endpoint-url  s3://ontapbucket1/
                           PRE dbpedia_csv/
2023-02-16 19:19:27      26774 AUTHORS
2023-02-16 19:19:27      72727 NEWS
2023-02-16 19:19:27       4493 README
2023-02-16 19:19:27       2825 deprecated.txt
2023-02-16 19:19:27       1590 getopt-parse.bash
2023-02-16 19:19:27       2245 getopt-parse.tcsh
sh-4.2$
```

验证从Jupyter笔记本电脑学习机器的情况

以下验证使用以下SageMaker BlazingText示例、通过文本分类提供机器学习构建、培训和部署模型：

1. 安装boto3和SageMaker软件包。

```
In [1]:  pip install --upgrade boto3 sagemaker
```

输出：

```
Looking in indexes: https://pypi.org/simple,
https://pip.repos.neuron.amazo naws.com
Requirement already satisfied: boto3 in /home/ec2-
user/anaconda3/envs/pytho n3/lib/python3.10/site-packages (1.26.44)
Collecting boto3
  Downloading boto3-1.26.72-py3-none-any.whl (132 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
132.7/132.7 kB 14.6 MB/s eta 0: 00:00
Requirement already satisfied: sagemaker in /home/ec2-
user/anaconda3/envs/p ython3/lib/python3.10/site-packages (2.127.0)
Collecting sagemaker
  Downloading sagemaker-2.132.0.tar.gz (668 kB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
668.0/668.0 kB 12.3 MB/s eta 0:
00:0000:01
  Preparing metadata (setup.py) ... done
Collecting botocore<1.30.0,>=1.29.72
  Downloading botocore-1.29.72-py3-none-any.whl (10.4 MB)
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━
10.4/10.4 MB 44.3 MB/s eta 0: 00:0000:010:01
Requirement already satisfied: s3transfer<0.7.0,>=0.6.0 in /home/ec2-
user/a naconda3/envs/python3/lib/python3.10/site-packages (from boto3)
(0.6.0)
```

```
Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /home/ec2-
user/ana conda3/envs/python3/lib/python3.10/site-packages (from boto3)
(0.10.0)
Requirement already satisfied: attrs<23,>=20.3.0 in /home/ec2-
user/anaconda
3/envs/python3/lib/python3.10/site-packages (from sagemaker) (22.1.0)
Requirement already satisfied: google-pasta in /home/ec2-
user/anaconda3/env s/python3/lib/python3.10/site-packages (from
sagemaker) (0.2.0)
Requirement already satisfied: numpy<2.0,>=1.9.0 in /home/ec2-
user/anaconda
3/envs/python3/lib/python3.10/site-packages (from sagemaker) (1.22.4)
Requirement already satisfied: protobuf<4.0,>=3.1 in /home/ec2-
user/anacond a3/envs/python3/lib/python3.10/site-packages (from
sagemaker) (3.20.3)
Requirement already satisfied: protobuf3-to-dict<1.0,>=0.1.5 in
/home/ec2-u ser/anaconda3/envs/python3/lib/python3.10/site-packages
(from sagemaker)
(0.1.5)
Requirement already satisfied: smdebug_rulesconfig==1.0.1 in /home/ec2-
use r/anaconda3/envs/python3/lib/python3.10/site-packages (from
sagemaker) (1.
0.1) Requirement already satisfied: importlib-metadata<5.0,>=1.4.0 in
/home/ec2user/anaconda3/envs/python3/lib/python3.10/site-packages (from
sagemaker)
(4.13.0)
Requirement already satisfied: packaging>=20.0 in /home/ec2-
user/anaconda3/ envs/python3/lib/python3.10/site-packages (from
sagemaker) (21.3)
Requirement already satisfied: pandas in /home/ec2-
user/anaconda3/envs/pyth on3/lib/python3.10/site-packages (from
sagemaker) (1.5.1)
Requirement already satisfied: pathos in /home/ec2-
user/anaconda3/envs/pyth on3/lib/python3.10/site-packages (from
sagemaker) (0.3.0)
Requirement already satisfied: schema in /home/ec2-
user/anaconda3/envs/pyth on3/lib/python3.10/site-packages (from
sagemaker) (0.7.5) Requirement already satisfied: python-
dateutil<3.0.0,>=2.1 in /home/ec2-use
r/anaconda3/envs/python3/lib/python3.10/site-packages (from
botocore<1.30.
0,>=1.29.72->boto3) (2.8.2)
Requirement already satisfied: urllib3<1.27,>=1.25.4 in /home/ec2-
user/anac onda3/envs/python3/lib/python3.10/site-packages (from
botocore<1.30.0,>=1.2
9.72->boto3) (1.26.8) Requirement already satisfied: zipp>=0.5 in
```

```
/home/ec2-user/anaconda3/envs/p ython3/lib/python3.10/site-packages
(from importlib-metadata<5.0,>=1.4.0->s agemaker) (3.10.0)
Requirement already satisfied: pyparsing!=3.0.5,>=2.0.2 in /home/ec2-
user/a naconda3/envs/python3/lib/python3.10/site-packages (from
packaging>=20.0->s agemaker) (3.0.9)
Requirement already satisfied: six in /home/ec2-
user/anaconda3/envs/python
3/lib/python3.10/site-packages (from protobuf3-to-dict<1.0,>=0.1.5-
>sagemak er) (1.16.0)
Requirement already satisfied: pytz>=2020.1 in /home/ec2-
user/anaconda3/env s/python3/lib/python3.10/site-packages (from pandas-
>sagemaker) (2022.5)
Requirement already satisfied: ppft>=1.7.6.6 in /home/ec2-
user/anaconda3/en vs/python3/lib/python3.10/site-packages (from pathos-
>sagemaker) (1.7.6.6) Requirement already satisfied:
multiprocess>=0.70.14 in /home/ec2-user/anac
onda3/envs/python3/lib/python3.10/site-packages (from pathos->sagemaker)
(0.70.14)
Requirement already satisfied: dill>=0.3.6 in /home/ec2-
user/anaconda3/env s/python3/lib/python3.10/site-packages (from pathos-
>sagemaker) (0.3.6)
Requirement already satisfied: pox>=0.3.2 in /home/ec2-
user/anaconda3/envs/ python3/lib/python3.10/site-packages (from pathos-
>sagemaker) (0.3.2) Requirement already satisfied: contextlib2>=0.5.5 in
/home/ec2-user/anacond a3/envs/python3/lib/python3.10/site-packages
(from schema->sagemaker) (21.
6.0) Building wheels for collected packages: sagemaker
  Building wheel for sagemaker (setup.py) ... done
  Created wheel for sagemaker: filename=sagemaker-2.132.0-py2.py3-none-
any. whl size=905449
sha256=f6100a5dc95627f2e2a49824e38f0481459a27805ee19b5a06ec
83db0252fd41
  Stored in directory: /home/ec2-
user/.cache/pip/wheels/60/41/b6/482e7ab096
520df034fbf2dddd244a1d7ba0681b27ef45aa61
Successfully built sagemaker
Installing collected packages: botocore, boto3, sagemaker
  Attempting uninstall: botocore     Found existing installation:
botocore 1.24.19
    Uninstalling botocore-1.24.19:       Successfully uninstalled
botocore-1.24.19
  Attempting uninstall: boto3     Found existing installation: boto3
1.26.44
    Uninstalling boto3-1.26.44:
      Successfully uninstalled boto3-1.26.44
  Attempting uninstall: sagemaker     Found existing installation:
```

```
sagemaker 2.127.0
    Uninstalling sagemaker-2.127.0:
      Successfully uninstalled sagemaker-2.127.0
ERROR: pip's dependency resolver does not currently take into account
all t he packages that are installed. This behaviour is the source of
the followi ng dependency conflicts.
awscli 1.27.44 requires botocore==1.29.44, but you have botocore 1.29.72
wh ich is incompatible.
aiobotocore 2.0.1 requires botocore<1.22.9,>=1.22.8, but you have
botocore 1.29.72 which is incompatible. Successfully installed boto3-
1.26.72 botocore-1.29.72 sagemaker-2.132.0 Note: you may need to restart
the kernel to use updated packages.
```

2. 在以下步骤中、将显示数据 (`dbpedia_csv`)下载 `ontapbucket1` 机器学习中使用的Jupyter笔记本实例。

```
In [2]: import sagemaker
In [3]: from sagemaker import get_execution_role
In [4]:
import json
import boto3
sess = sagemaker.Session()
role = get_execution_role()
print(role)
bucket = "ontapbucket1"
print(bucket)
sess.s3_client = boto3.client('s3',region_name='',aws_access_key_id =
'0ZNAX21JW5Q8AP80CQ2E',  aws_secret_access_key =
'PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr',
                              use_ssl = False, endpoint_url =
'http://172.30.10.41',

config=boto3.session.Config(signature_version='s3v4',
s3={'addressing_style':'path'}) )
sess.s3_resource = boto3.resource('s3',region_name='',aws_access_key_id
= '0ZNAX21JW5Q8AP80CQ2E', aws_secret_access_key =
'PpLs4gA9K0_2gPhuykkp014gBjcC9Rbi3QDX_6rr',
                              use_ssl = False, endpoint_url =
'http://172.30.10.41',

config=boto3.session.Config(signature_version='s3v4',
s3={'addressing_style':'path'}) )
prefix = "blazingtext/supervised"
import os
my_bucket = sess.s3_resource.Bucket(bucket)
my_bucket = sess.s3_resource.Bucket(bucket)
#os.mkdir('dbpedia_csv')
for s3_object in my_bucket.objects.all():
    filename = s3_object.key
#    print(filename)
#    print(s3_object.key)
    my_bucket.download_file(s3_object.key, filename)
```

3. 以下代码将创建从整数索引到类标签的映射、用于在推理期间检索实际类名称。

```
index_to_label = {}
with open("dbpedia_csv/classes.txt") as f:
    for i,label in enumerate(f.readlines()):
        index_to_label[str(i + 1)] = label.strip()
```

输出将列出中的文件和文件夹 `ontapbucket1` 用于AWS SageMaker机器学习验证的数据分段。

```
arn:aws:iam::210811600188:role/SageMakerFullRole ontapbucket1
AUTHORS
AUTHORS
NEWS
NEWS
README README
dbpedia_csv/classes.txt dbpedia_csv/classes.txt dbpedia_csv/readme.txt
dbpedia_csv/readme.txt dbpedia_csv/test.csv dbpedia_csv/test.csv
dbpedia_csv/train.csv dbpedia_csv/train.csv deprecated.txt
deprecated.txt getopt-parse.bash getopt-parse.bash getopt-parse.tcsh
getopt-parse.tcsh
In [5]: ls
AUTHORS       deprecated.txt     getopt-parse.tcsh  NEWS
Untitled.ipynb dbpedia_csv/  getopt-parse.bash  lost+found/
README
In [6]: ls -l dbpedia_csv
total 191344
-rw-rw-r-- 1 ec2-user ec2-user       146 Feb 16 19:43 classes.txt
-rw-rw-r-- 1 ec2-user ec2-user      1758 Feb 16 19:43 readme.txt
-rw-rw-r-- 1 ec2-user ec2-user  21775285 Feb 16 19:43 test.csv
-rw-rw-r-- 1 ec2-user ec2-user 174148970 Feb 16 19:43 train.csv
```

4. 启动数据预处理阶段、将训练数据预处理为空格分隔的令牌化文本格式、BlazingText算法和Nultk库可以使用该格式来标记DBPedia数据集的输入语句。下载nltk令牌程序和其他库。。 `transform_instance` 并行应用于每个数据实例时、会使用Python多处理模块。

```
ln [7]: from random import shuffle
import multiprocessing
from multiprocessing import Pool
import csv
import nltk
nltk.download("punkt")
def transform_instance(row):
    cur_row = []
    label ="__label__" + index_to_label [row[0]] # Prefix the index-ed
label with __label__
    cur_row.append (label)
    cur_row.extend(nltk.word_tokenize(row[1].lower ()))
    cur_row.extend(nltk.word_tokenize(row[2].lower ()))
    return cur_row
def preprocess(input_file, output_file, keep=1):
    all_rows = []
    with open(input_file,"r") as csvinfile:
```

```
        csv_reader = csv.reader(csvinfile, delimiter=",")
        for row in csv_reader:
            all_rows.append(row)
    shuffle(all_rows)
    all_rows = all_rows[: int(keep * len(all_rows))]
    pool = Pool(processes=multiprocessing.cpu_count())
    transformed_rows = pool.map(transform_instance, all_rows)
    pool.close()
    pool. join()
    with open(output_file, "w") as csvoutfile:
        csv_writer = csv.writer (csvoutfile, delimiter=" ",
lineterminator="\n")
        csv_writer.writerows (transformed_rows)

# Preparing the training dataset
# since preprocessing the whole dataset might take a couple of minutes,
# we keep 20% of the training dataset for this demo.
# Set keep to 1 if you want to use the complete dataset
preprocess("dbpedia_csv/train.csv","dbpedia.train", keep=0.2)
# Preparing the validation dataset
preprocess("dbpedia_csv/test.csv","dbpedia.validation")
sess = sagemaker.Session()
role = get_execution_role()
print (role) # This is the role that sageMaker would use to leverage Aws
resources (S3,  Cloudwatch) on your behalf
bucket = sess.default_bucket() # Replace with your own bucket name if
needed
print("default Bucket::: ")
print(bucket)
```

输出：

```
[nltk_data] Downloading package punkt to /home/ec2-user/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
arn:aws:iam::210811600188:role/SageMakerFullRole default Bucket:::
sagemaker-us-east-1-210811600188
```

5. 将格式化的训练数据集上传到S3、以便SageMaker可以使用它来执行训练作业。然后、使用Python SDK将两个文件上传到存储分段和前缀位置。

```
ln [8]: %%time
train_channel = prefix + "/train"
validation_channel = prefix + "/validation"
sess.upload_data(path="dbpedia.train", bucket=bucket,
key_prefix=train_channel)
sess.upload_data(path="dbpedia.validation", bucket=bucket,
key_prefix=validation_channel)
s3_train_data = "s3://{}/{}".format(bucket, train_channel)
s3_validation_data = "s3://{}/{}".format(bucket, validation_channel)
```

输出：

```
CPU times: user 546 ms, sys: 163 ms, total: 709 ms
Wall time: 1.32 s
```

6. 在S3上设置加载模型项目的输出位置、以便项目可以作为算法训练作业的输出。创建 `sageMaker.estimator.Estimator` 用于启动培训作业的对象。

```
In [9]: s3_output_location = "s3://{}/{}/output".format(bucket, prefix)
In [10]: region_name = boto3.Session().region_name
In [11]: container =
sagemaker.amazon.amazon_estimator.get_image_uri(region_name,
"blazingtext","latest")
print("Using SageMaker BlazingText container: {} ({})".format(container,
region_name))
```

输出：

```
The method get_image_uri has been renamed in sagemaker>=2.
See: https://sagemaker.readthedocs.io/en/stable/v2.html for details.
Defaulting to the only supported framework/algorithm version: 1.
Ignoring f ramework/algorithm version: latest.
Using SageMaker BlazingText container: 811284229777.dkr.ecr.us-east-
1.amazo naws.com/blazingtext:1 (us-east-1)
```

7. 定义SageMaker `Estrimator` 通过资源配置和超参数、在c4.4xlarge实例上使用受监控模式在DBPedia数据集上训练文本分类。

```
In [12]: bt_model = sagemaker.estimator.Estimator(
container,
role,
instance_count=1,
instance_type="ml.c4.4xlarge",
volume_size=30,
max_run=360000,
input_mode="File",
output_path=s3_output_location,
hyperparameters={
        "mode": "supervised",
        "epochs": 1,
        "min_count": 2,
        "learning_rate": 0.05,
        "vector_dim": 10,
        "early_stopping": True,
        "patience": 4,
        "min_epochs": 5,
        "word_ngrams": 2,
 },
     )
```

8. 准备数据通道和算法之间的握手。为此、请创建 `sagemaker.session.s3_input` 数据通道中的对象、并将其保留在词典中、以供算法使用。

```
ln [13]: train_data = sagemaker.inputs.TrainingInput(
    s3_train_data,
    distribution="FullyReplicated",
    content_type="text/plain",
    s3_data_type="S3Prefix",
)
validation_data = sagemaker.inputs.TrainingInput(
    s3_validation_data,
    distribution="FullyReplicated",
    content_type="text/plain",
    s3_data_type="S3Prefix",
)
data_channels = {"train": train_data, "validation": validation_data}
```

9. 作业完成后、将显示作业完成消息。您可以在设置为的S3存储分段中找到经过培训的型号 `output_path` 在估算器中。

```
ln [14]: bt_model.fit(inputs=data_channels, logs=True)
```

输出：

```
INFO:sagemaker:Creating training-job with name: blazingtext-2023-02-16-
20-3
7-30-748
2023-02-16 20:37:30 Starting - Starting the training job......
2023-02-16 20:38:09 Starting - Preparing the instances for
training......
2023-02-16 20:39:24 Downloading - Downloading input data
2023-02-16 20:39:24 Training - Training image download completed.
Training in progress... Arguments: train
[02/16/2023 20:39:41 WARNING 140279908747072] Loggers have already been
set up. [02/16/2023 20:39:41 WARNING 140279908747072] Loggers have
already been set up.
[02/16/2023 20:39:41 INFO 140279908747072] nvidia-smi took:
0.0251793861389
16016 secs to identify 0 gpus
[02/16/2023 20:39:41 INFO 140279908747072] Running single machine CPU
Blazi ngText training using supervised mode.
Number of CPU sockets found in instance is  1
[02/16/2023 20:39:41 INFO 140279908747072] Processing
/opt/ml/input/data/tr ain/dbpedia.train . File size: 35.0693244934082 MB
[02/16/2023 20:39:41 INFO 140279908747072] Processing
/opt/ml/input/data/va lidation/dbpedia.validation . File size:
21.887572288513184 MB
Read 6M words
Number of words:  149301
Loading validation data from
/opt/ml/input/data/validation/dbpedia.validati on
Loaded validation data.
-------------- End of epoch: 1 ##### Alpha: 0.0000  Progress: 100.00%
Million Words/sec: 10.39 ##### Training finished.
Average throughput in Million words/sec: 10.39
Total training time in seconds: 0.60
#train_accuracy: 0.7223
Number of train examples: 112000
#validation_accuracy: 0.7205
Number of validation examples: 70000
2023-02-16 20:39:55 Uploading - Uploading generated training model
2023-02-16 20:40:11 Completed - Training job completed
Training seconds: 68
Billable seconds: 68
```

10. 培训完成后、将经过培训的模型部署为Amazon SageMaker实时托管端点、以进行预测。

```
In [15]: from sagemaker.serializers import JSONSerializer
 text_classifier = bt_model.deploy(
     initial_instance_count=1, instance_type="ml.m4.xlarge",
serializer=JSONS
)
```

输出:

```
INFO:sagemaker:Creating model with name: blazingtext-2023-02-16-20-41-
33-10
0
INFO:sagemaker:Creating endpoint-config with name blazingtext-2023-02-
16-20
-41-33-100
INFO:sagemaker:Creating endpoint with name blazingtext-2023-02-16-20-41-
33-
100
-------!
```

```
In [16]: sentences = [
    "Convair was an american aircraft manufacturing company which later
expanded into rockets and spacecraft.",
       "Berwick secondary college is situated in the outer melbourne
metropolitan suburb of berwick .",
]
# using the same nltk tokenizer that we used during data preparation for
training
tokenized_sentences = [" ".join(nltk.word_tokenize(sent)) for sent in
sentences]
payload = {"instances": tokenized_sentences} response =
text_classifier.predict(payload)
predictions = json.loads(response)
print(json.dumps(predictions, indent=2))
```

```
[
  {
    "label": [
      "__label__Artist"
    ],
    "prob": [
      0.4090951681137085
    ]
  },
  {
    "label": [
      "__label__EducationalInstitution"
    ],
    "prob": [
      0.49466073513031006
    ]
  }
]
```

11. 默认情况下、模型返回一个概率最高的预测。以检索顶部 `k` 预测、设置 `k` 在配置文件中。

```
In [17]: payload = {"instances": tokenized_sentences, "configuration":
{"k": 2}}
 response = text_classifier.predict(payload)

 predictions = json.loads(response)
 print(json.dumps(predictions, indent=2))
```

```
[
  {
    "label": [
      "__label__Artist",
      "__label__MeanOfTransportation"
    ],
    "prob": [
      0.4090951681137085,
      0.26930734515190125
    ]
  },
  {
    "label": [
      "__label__EducationalInstitution",
      "__label__Building"
    ],
    "prob": [
      0.49466073513031006,
      0.15817692875862122
    ]
  }
]
```

12. 请先删除端点、然后再关闭此笔记本。

```
In [18]: sess.delete_endpoint(text_classifier.endpoint)
WARNING:sagemaker.deprecations:The endpoint attribute has been renamed
in s agemaker>=2.
See: https://sagemaker.readthedocs.io/en/stable/v2.html for details.
INFO:sagemaker:Deleting endpoint with name: blazingtext-2023-02-16-20-
41-33
-100
```

结论

根据此验证、数据科学家和工程师可以通过NetApp Cloud Volumes ONTAP 的S3存储分段从AWS SageMaker Jupyter笔记本电脑访问NFS数据。这种方法可以轻松地从NFS和S3访问和共享相同的数据、而无需额外的软件。

从何处查找追加信息

要了解有关本文档中所述信息的更多信息，请查看以下文档和 / 或网站：

- 使用SageMaker BlazingText进行文本分类

  "https://sagemaker-examples.readthedocs.io/en/latest/introduction_to_amazon_algorithms/blazingtext_text_classification_dbpedia/blazingtext_text_classification_dbpedia.html"

- S3 对象存储的 ONTAP 版本支持

  "https://docs.netapp.com/us-en/ontap/s3-config/ontap-version-support-s3-concept.html"

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